Quantitative Research

Part 3: Statistical Inference

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HSBI

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The lecture at a glance

	Part	Content
	1 – Descriptive Statistics	Basic concepts, levels of measurement, frequencies, measures of central tendency and dispersion
	2 – Graphical Representations	Empirical CDF, histogram, boxplot
▶	**3 – Statistical Inference**	**Confidence intervals, hypothesis tests, p-values, one-sided tests, types of error**
	4 – Regression	Correlation, simple and multiple linear regression, inference for regression, diagnostics
	5 – Classification	Logistic regression, classification rules, confusion matrix, ROC/AUC, Bayes classifiers

All analyses are carried out in **R** (with RStudio); every deck comes with an R script that reproduces its numbers.

Contents – Part 3

1. Confidence Intervals
2. Hypothesis Tests
3. The p-Value
4. One-Sided Tests
5. Types of Error
6. Summary

The common thread of this part

One question runs through every section

We draw a **sample** and want to learn something about the **population** from it – while being *honest* about the remaining uncertainty.

Two tools, one idea:

- **Confidence interval** – "Which values of μ are *compatible* with the data?"
- **Hypothesis test** – "Is a specific value μ_0 compatible – or do the data *reject* it?"

One running story:

The dairy

A machine fills milk. Is it working correctly? We measure, estimate and test – and follow this example throughout the entire part.

Learning objectives – Part 3

After this part you will be able to ...

- compute and correctly interpret a **confidence interval** for μ,
- carry out a **hypothesis test** (known/unknown variance, one-/two-sided),
- compute the p-**value** and read it as a measure of evidence,
- distinguish **Type I and Type II errors** as well as the **power** of a test,
- read **critical values** from the z- and t-tables and link them to the p-value,
- implement every procedure in `R`.

Key takeaway

Common thread: infer from a sample – and report the uncertainty honestly.

List of symbols (1/2)

Symbol	Meaning	Symbol	Meaning
μ	Expected value	n	Sample size
σ^2	Variance	X_i	Sample variable
σ	Standard deviation	x_i	Observed value
$\bar{X}$	Sample mean	μ_0	Hypothesized value
$\bar{x}$	Realization	s^2	Sample variance
s	Sample std. dev.	H_0	Null hypothesis
H_1	Alternative hypothesis	α	Type I error prob.
β	Type II error prob.	$1-\beta$	Power

Convention: $N(\mu, \sigma^2)$ – the second argument is the **variance** (not the standard deviation).

List of symbols (2/2)

Symbol	Meaning	Symbol	Meaning
Z_n	Test stat., known var.	T_n	Test stat., unknown var.
$z_{1-\alpha/2}$	Two-sided quantile, $N(0,1)$	$t_{1-\alpha/2,n-1}$	Two-sided quantile, t_{n-1}
$z_{1-\alpha}$	One-sided quantile, $N(0,1)$	$t_{1-\alpha,n-1}$	One-sided quantile, t_{n-1}
$\Phi(\cdot)$	CDF of $N(0,1)$	$F_{t,n-1}$	CDF of t_{n-1}
p	p-value	CI	Confidence interval
α	Significance level	$1-\alpha$	Confidence level
β	Type II error prob.	$1-\beta$	Power

Overview

1 Confidence Intervals
- Motivation and intuition
- Known variance
- Length of the confidence interval
- Unknown variance
- Exercises

2 Hypothesis Tests

3 The p-Value

4 One-Sided Tests

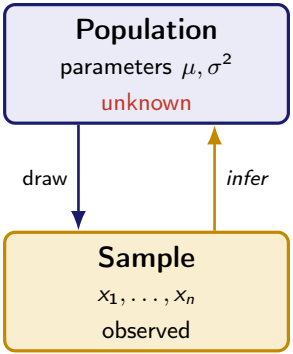
5 Types of Error

6 Summary

Why inferential statistics?

The starting question

We only know one **sample** – but want to say something about the **whole population**.



- **Descriptive:** describes the data at hand.
- **Inferential:** infers *back* from the sample to the population.
- This inference is *uncertain* – we must report the uncertainty.

A single estimate is not enough

Intuition

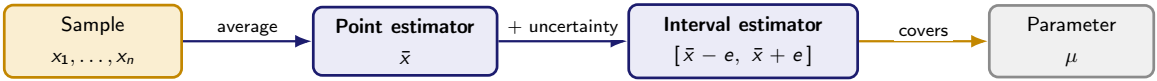
From the sample we compute the mean $\bar{x}$. But it almost **never** hits the true value μ exactly – a second sample would give a different value.

- Instead of a single point we report a **range** of values that are *compatible with the data*.
- The more precise our estimate, the *narrower* this range.

Key takeaway

A **confidence interval** reports the estimate *together with its uncertainty* – a range of plausible values for μ. (It is *not* a probability statement about where μ lies – see the interpretation slides.)

From the sample to the interval

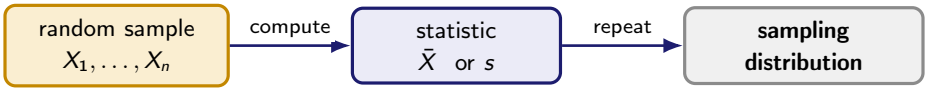


- The point estimator $\bar{x}$ is the *best single estimate*.
- The *margin of error* $e = z_{1-\alpha/2}\,\sigma/\sqrt{n}$ (half the interval's width) turns it into an interval – built so that the *procedure* covers the fixed true μ in $(1-\alpha)$ of samples.

Estimators are random variables

The sample $X_1, \ldots, X_n$ is **random**, so *anything* computed from it – $\bar{X}$, s, Z_n, T_n – is itself a **random variable**.

- New sample $\Rightarrow$ a new $\bar{x}$ *and* a new s.
- Its distribution over *all possible samples* is the **sampling distribution** – so $\bar{X}$ and s each have their own.



Key takeaway

A confidence interval or test is built on the *sampling distribution* of an estimator – never on a single number alone.

The standard error: how much $\bar{X}$ wobbles

Definition

The **standard error** (SE) is the *standard deviation of the sampling distribution* of an estimator – the spread of $\bar{X}$ from sample to sample, *not* the spread of the raw data.

$$\mathrm{SE}(\bar{X}) = \frac{\sigma}{\sqrt{n}} \quad \Big(\text{or } \tfrac{s}{\sqrt{n}} \text{ when } \sigma \text{ is estimated}\Big).$$

- It is the natural **ruler** for inference: Z_n and T_n count *how many standard errors* $\bar{x}$ sits away from μ_0.
- Larger $n \Rightarrow$ smaller SE $\Rightarrow$ narrower CI and sharper test.

Key takeaway

Standard *deviation* = spread of the data; standard *error* = spread of the estimate.

Why divide by the standard error?

The raw deviation is not enough

A sample mean $\bar{x} = 8$ lies 2 units from a reference value $\mu_0 = 10$. Whether that gap is *small or large* depends entirely on the standard error – how much $\bar{X}$ wobbles.

Scenario	$\mathrm{SE} = \sigma/\sqrt{n}$	$z = \dfrac{8-10}{\mathrm{SE}}$	Reading
large spread, small n $(\sigma = 6,\ n = 9)$	2.0	-1.0	ordinary
small spread, large n $(\sigma = 4,\ n = 100)$	0.4	-5.0	far out

Key takeaway

Same raw deviation, opposite readings. Dividing by the SE turns it into a *universal* ruler – "how many standard errors" – comparable across samples and units.

How we read probability: the frequentist view

Fixed vs. random

- Parameters μ, σ are **fixed** (unknown) constants – not random.
- The data – and all we build from them ($\bar{X}$, s, the CI bounds, Z_n, p) – are **random**.

What "probability" means here

The long-run *relative frequency* of an event over (hypothetically) very many repeated samples.

Key takeaway

"95% confidence" and "5% significance" describe the **procedure across repeated samples** – not one particular interval and not one particular dataset.

Our story: filling milk

Example with known variance

- A dairy fills milk (label: $1\,\text{l} = 1000\,\text{ml}$).
- Machine with **known** variance $\sigma^2 = 9\,\text{ml}^2$, i.e. $\sigma = 3\,\text{ml}$.
- After servicing: is it still working correctly?
- Sample $n = 9$ (in ml):

1005, 1001, 997, 1000, 1001, 998, 999, 999, 1009

Key figures

$$\bar{x} = 1001\,\text{ml}$$

$$1 - \alpha = 0.95 \;\Rightarrow\; \alpha = 0.05$$

Goal: confidence interval for the true mean μ.

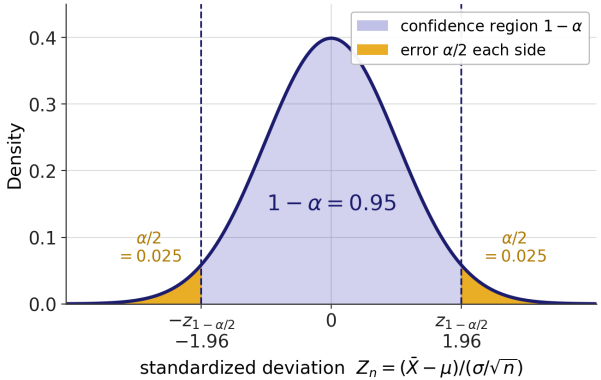
How does the mean fluctuate? – (1) intuition

Thought experiment

Imagine we draw the sample *over and over again*. Each time we get a slightly different $\bar{x}$.

- These $\bar{x}$-values scatter around the true value μ.
- But they scatter **more tightly** than the individual values – the average evens out outliers.
- The standard error shrinks like $1/\sqrt{n}$: four times as much data $\Rightarrow$ half the spread *of the sample mean*.
- Reason: variances *add* for independent values, so the spread grows like $\sqrt{n}$ for the *sum* but shrinks like $1/\sqrt{n}$ once we *average*. (For non-normal data the **Central Limit Theorem** makes $\bar{X}$ approximately normal for large n.)

How does the mean fluctuate? – (2) graphically



Distribution of the standardized statistic $Z_n \sim N(0,1)$: the central (blue) region holds 95 % of the probability – in 95 % of samples Z_n lands here; 2.5 % in each tail.

How does the mean fluctuate? – (3) formally

Sampling distribution of the sample mean

For $X_i \sim N(\mu, \sigma^2)$ i.i.d., the *sampling distribution* of the random variable $\bar{X}$ is

$$\bar{X} \sim N\Big(\mu,\ \tfrac{\sigma^2}{n}\Big) \quad \Longrightarrow \quad Z_n := \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \sim N(0,1).$$

Why σ^2/n? Averaging n independent draws divides the variance by n ($\mathrm{Var}(\bar{X}) = \sigma^2/n$), so the standard error is $\sigma/\sqrt{n}$; Z_n is just $\bar{X}$ centred and scaled to $N(0,1)$.

Two-sided probability statement

$$P\big(-z_{1-\alpha/2} \le Z_n \le z_{1-\alpha/2}\big) = 1 - \alpha$$

In words: With probability $1 - \alpha$, Z_n lies between the two quantiles $\pm z_{1-\alpha/2}$.

Derivation (1/2): rearrange step by step

Start from the pivot $Z_n = \dfrac{\bar{X} - \mu}{\sigma/\sqrt{n}} \sim N(0,1)$, which lies within $\pm z_{1-\alpha/2}$ with probability $1-\alpha$:

$$1-\alpha = P\Big(-z_{1-\alpha/2} \;\le\; \tfrac{\bar{X}-\mu}{\sigma/\sqrt{n}} \;\le\; z_{1-\alpha/2}\Big) \qquad \triangleright \text{ starting point}$$

$$= P\Big(-z_{1-\alpha/2}\tfrac{\sigma}{\sqrt{n}} \;\le\; \bar{X} - \mu \;\le\; z_{1-\alpha/2}\tfrac{\sigma}{\sqrt{n}}\Big) \qquad \triangleright \text{ multiply all 3 parts by } \tfrac{\sigma}{\sqrt{n}} > 0$$

$$= P\Big(-\bar{X} - z_{1-\alpha/2}\tfrac{\sigma}{\sqrt{n}} \;\le\; -\mu \;\le\; -\bar{X} + z_{1-\alpha/2}\tfrac{\sigma}{\sqrt{n}}\Big) \qquad \triangleright \text{ subtract } \bar{X} \text{ everywhere}$$

In words: Multiplying by a *positive* number keeps the "$\le$" signs; we only used equivalence steps, so the probability stays exactly $1-\alpha$.

Derivation (2/2): from $-\mu$ to the interval

$$1-\alpha = P\Big(\bar{X} + z_{1-\alpha/2}\tfrac{\sigma}{\sqrt{n}} \;\geq\; \mu \;\geq\; \bar{X} - z_{1-\alpha/2}\tfrac{\sigma}{\sqrt{n}}\Big) \qquad \triangleright \text{ multiply by } (-1): \text{ signs flip}$$

$$= P\Big(\bar{X} - z_{1-\alpha/2}\tfrac{\sigma}{\sqrt{n}} \;\leq\; \mu \;\leq\; \bar{X} + z_{1-\alpha/2}\tfrac{\sigma}{\sqrt{n}}\Big) \qquad \triangleright \text{ just read it left-to-right}$$

Key takeaway

Confidence interval with **known** variance:

$$\mathrm{CI}_{1-\alpha} = \left[\bar{X} - z_{1-\alpha/2}\frac{\sigma}{\sqrt{n}},\; \bar{X} + z_{1-\alpha/2}\frac{\sigma}{\sqrt{n}}\right]$$

Multiplying an inequality by -1 **reverses** "$\leq$" into "$\geq$" – that is the only subtle step. What is random are the *bounds* (they contain $\bar{X}$); μ itself is a fixed unknown.

The formula in words

$$\text{CI} = \Big[\underbrace{\bar{X}}_{\text{estimate}} \pm \underbrace{z_{1-\alpha/2}}_{\substack{\text{critical} \\ \text{value}}} \cdot \underbrace{\frac{\sigma}{\sqrt{n}}}_{\text{standard error}} \Big]$$

- **Centre** $\bar{X}$: our best estimate of μ.
- **Standard error** $\sigma/\sqrt{n}$: how strongly $\bar{X}$ fluctuates.
- **Critical value** $z_{1-\alpha/2}$: how many standard errors we go to the left/right (1.96 at 95 %).

Milk example: calculation

Given: $\bar{x} = 1001$, $\sigma = 3$, $n = 9$, $\alpha = 0.05$, $z_{0.975} = 1.96$.

$$\text{margin of error } e = z_{1-\alpha/2} \cdot \frac{\sigma}{\sqrt{n}} = 1.96 \cdot \frac{3}{\sqrt{9}} = 1.96$$

$$\text{lower bound} = 1001 - 1.96 = 999.04$$

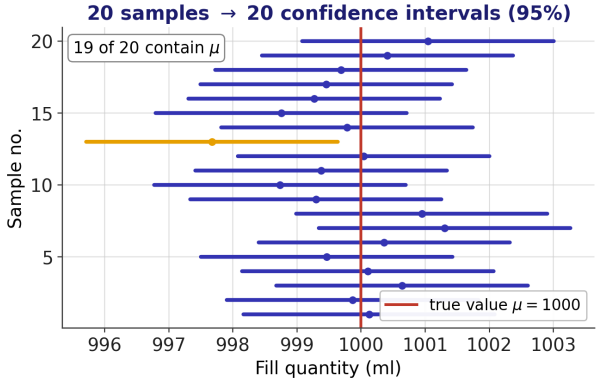
$$\text{upper bound} = 1001 + 1.96 = 1002.96$$

Result

$$\text{CI}_{0.95} = \big[\, 999.04\,;\, 1002.96 \,\big] \text{ ml}$$

The hypothesized target value $\mu_0 = 1000$ lies *inside* the interval $\Rightarrow$ no evidence against a correctly working machine.

What does "95 % confidence" mean?



Of 20 computed 95 % intervals, on average ≈ 19 cover the true value μ (red line). The confidence *level* refers to the **procedure** across many samples.

A common misconception

Careful with the interpretation

"With 95 % probability μ lies in *this particular* interval" is, strictly speaking, **wrong**.

- μ is a fixed (unknown) number – it either is or is not inside the particular interval.
- What is random are the *bounds* of the interval, not μ.
- Correct: "The **procedure** yields, in 95 % of samples, an interval that covers μ."

Three correct ways to read a confidence interval

1. **Coverage (the procedure).** The method produces intervals that contain the fixed true μ in a fraction $(1 - \alpha)$ of all samples.
2. **Compatibility.** The interval is the range of values for μ that are *consistent with the observed data*.
3. **Duality with the test.** It is exactly the set of null values μ_0 that a two-sided level-α test would *not* reject.

Key takeaway

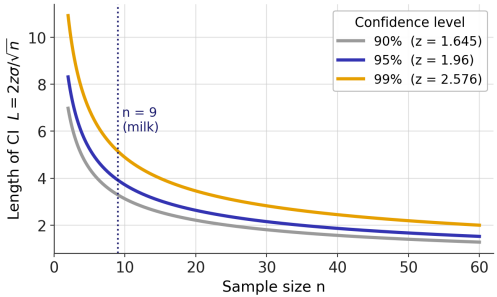
All three avoid the trap of putting a probability on *this one* interval: μ is fixed, the **interval** is what is random.

What does the length depend on?

Length

$$L = 2 \cdot z_{1-\alpha/2} \cdot \frac{\sigma}{\sqrt{n}}$$

- Higher confidence level $\Rightarrow$ larger $z_{1-\alpha/2}$ $\Rightarrow$ *longer*.
- Larger $n \Rightarrow$ *shorter* (with $1/\sqrt{n}$).
- Larger spread $\sigma \Rightarrow$ *longer*.



In words: More confidence *or* more precision – both require a larger sample.

Different confidence levels – milk

Confidence level	Quantile $z_{1-\alpha/2}$	CI for milk example
90 %	$z_{0.95} = 1.645$	[999.36 ; 1002.65]
95 %	$z_{0.975} = 1.96$	[999.04 ; 1002.96]
99 %	$z_{0.995} = 2.576$	[998.42 ; 1003.58]

Key takeaway

Higher confidence level $\Rightarrow$ wider interval $\Rightarrow$ less precise statement. Confidence and precision are a trade-off.

More realistic: the variance is unknown

Problem

In practice σ^2 is almost never known. We must *also estimate* it from the sample.

Estimator of the variance

$$s^2 = \frac{1}{n-1} \sum_{i=1}^{n} (X_i - \bar{X})^2$$

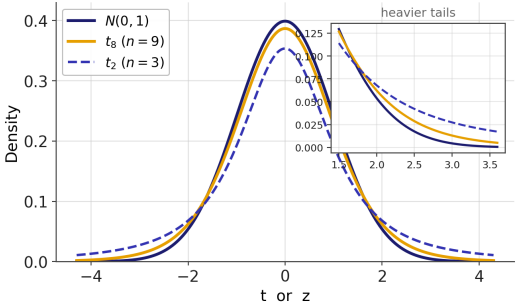
- We replace σ by s in the test statistic.
- This costs *accuracy* – and changes the distribution (see next slide).

Why the t-distribution? – intuitively and graphically

Standardized statistic (normal data)

$$T_n = \frac{\bar{X} - \mu}{s/\sqrt{n}} \sim t_{n-1}$$

- s is **itself a random variable**.
- T_n is a ratio of *two* random quantities $\Rightarrow$ **heavier tails**.
- For large n, s barely varies, so $t_{n-1} \to N(0,1)$.



Key takeaway

With unknown σ always use the t_{n-1}-distribution – its quantiles are larger, so the CI is wider.

Milk example with unknown variance

Confidence interval

$$\mathrm{CI} = \left[\bar{X} - t_{1-\alpha/2,\, n-1}\frac{s}{\sqrt{n}},\ \bar{X} + t_{1-\alpha/2,\, n-1}\frac{s}{\sqrt{n}}\right]$$

Data: $\bar{x} = 1001$, $n = 9$, $\alpha = 0.05$, $s = 3.7749$, $t_{0.975;\,8} = 2.3060$.

$$e = 2.3060 \cdot \frac{3.7749}{\sqrt{9}} = 2.9017$$

$$\mathrm{CI}_{0.95} = [\,1001 \pm 2.9017\,] = [\,998.10\,;\,1003.90\,]$$

Comparison

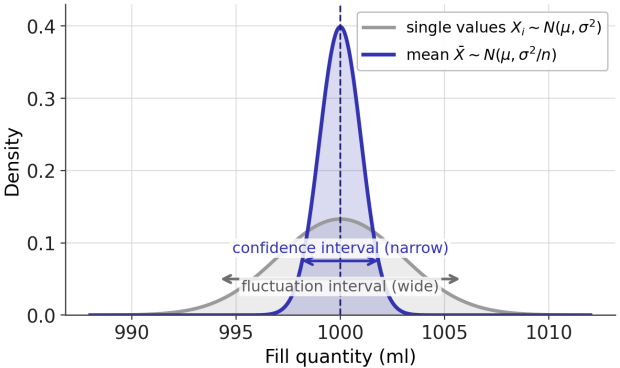
Wider than [999.04 ; 1002.96] (known variance) – the price for estimating σ.

Summary: known vs. unknown variance

	known σ	**unknown σ**
Spread used	σ	s (estimated)
Distribution	$N(0,1)$	t_{n-1}
Quantile (95 %, $n = 9$)	1.96	2.31
CI	narrower	wider

- For large n ($n \gtrsim 30$) the difference is small ($t_{n-1} \approx N(0,1)$).
- For small samples it is clearly noticeable.

Do not confuse: CI vs. fluctuation interval



- **Fluctuation interval:** spread of *individual* values X_i around μ – uses σ (wide).
- **Confidence interval:** uncertainty about μ – uses $\sigma/\sqrt{n}$ (narrow).

In words: Both intervals share the same centre but answer completely different questions.

Exercise – filling cream

Problem statement

A dairy fills cream into cups (label: 250 g). The variance of the machine is known: $\sigma^2 = 4\,\text{g}^2$. After servicing, $n = 16$ cups are sampled (in g):

$$251, 248, 248, 249, 250, 251, 247, 246, 252, 251, 248, 247, 246, 249, 248, 252$$

Confidence level $1 - \alpha = 0.95$.

Required: confidence interval for μ – (a) with known, (b) with unknown variance.

Solution (1/2) – cream, known variance

Rule: $\mathrm{CI} = \left[\bar{x} \pm z_{1-\alpha/2}\,\sigma/\sqrt{n}\right]$.
Intermediate values: $\bar{x} = 248.9375$, $\sigma = 2$, $n = 16$, $z_{0.975} = 1.96$.

$$e = 1.96 \cdot \frac{2}{\sqrt{16}} = 1.96 \cdot 0.5 = 0.98$$
$$\mathrm{CI}_{0.95} = [\,248.9375 \pm 0.98\,] = [\,\mathbf{247.96}\,;\,\mathbf{249.92}\,]$$

Interpretation

The target value $\mu_0 = 250$ lies *outside* the CI $\Rightarrow$ with known variance: indication that the machine is **not** working correctly.

Solution (2/2) – cream, unknown variance

Rule: as above, but $z \to t_{1-\alpha/2,\,n-1}$ and $\sigma \to s$.
Intermediate values: $s^2 = 4.0625 \Rightarrow s = 2.0156,\ t_{0.975;\,15} = 2.1314.$

$$e = 2.1314 \cdot \frac{2.0156}{\sqrt{16}} = 1.0740$$
$$\mathrm{CI}_{0.95} = [\,248.9375 \pm 1.0740\,] = [\,\mathbf{247.86}\,;\,\mathbf{250.01}\,]$$

Comparison

Now $\mu_0 = 250$ lies (just) *inside* the CI – the verdict turns out *differently* than with known variance. The wider t-interval is more cautious.

Summary and outlook

- Confidence intervals quantify the **uncertainty** of a point estimate.
- Known variance: quantiles of $N(0,1)$; unknown: of t_{n-1}.
- **Bridge to the test:** if the CI contains the hypothesized value μ_0, there is no evidence against $H_0 : \mu = \mu_0$ – otherwise there is.

Next section

We formalize this procedure as a **hypothesis test**.

Overview

1 Confidence Intervals

2 Hypothesis Tests
- Idea and hypotheses
- From CI to test
- Known variance
- Unknown variance
- Exercises
- Tests with R

3 The p-Value

4 One-Sided Tests

5 Types of Error

6 Summary

What is a hypothesis test? – intuition

The basic idea, as in court

An assumption (the *null hypothesis* H_0) is held true until the data **speak strongly enough against it**.

- H_0 = "innocent" (status quo).
- We reject H_0 only on *convincing* evidence.
- Little evidence $\Rightarrow$ acquittal for lack of evidence – not "proven innocent".

Milk

$H_0 : \mu = 1000$ – "the machine works correctly". Only a clear deviation from 1000 makes us doubt this.

The two hypotheses

Structure

$$H_0 : \mu = \mu_0 \qquad \text{vs.} \qquad H_1 : \mu \neq \mu_0$$

- We **always** argue under "H_0 is true".
- Why unequal? $H_0 : \mu = \mu_0$ fixes a *single* value, so the test statistic has *one* definite distribution we can compute with; H_1 is a whole *range*, with no single distribution.
- Two outcomes: *keep* H_0 or *reject* it.

Key takeaway

What you want to **establish** statistically goes into H_1.

Which hypothesis goes where?

Rule of thumb

The statement you want to *find evidence for* – the "more important" claim – is formulated as H_1, because only the Type I error (wrongly rejecting H_0) can be controlled via α.

You want evidence that ...	H_0	H_1
the supplier's fill weight *deviates* from the spec	$\mu = \mu_0$	$\mu \neq \mu_0$
the mean processing time *changed* after the update	$\mu = \mu_0$	$\mu \neq \mu_0$
the mean daily revenue *differs* from the plan figure	$\mu = \mu_0$	$\mu \neq \mu_0$

Bridge: confidence interval and test are the same

Testing with the CI

If μ_0 lies *inside* the confidence interval, $H_0 : \mu = \mu_0$ is **not** rejected – if it lies outside, it is.

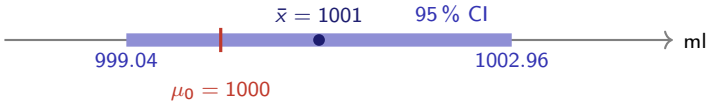
Equivalence – same algebra, two readings

"μ_0 inside the CI" means $|\bar{x} - \mu_0| \leq z_{1-\alpha/2}\,\sigma/\sqrt{n}$, which rearranges to $|Z_n| \leq z_{1-\alpha/2}$ – exactly the non-rejection condition. So a test (level α) and a CI (level $1-\alpha$) **always** agree.

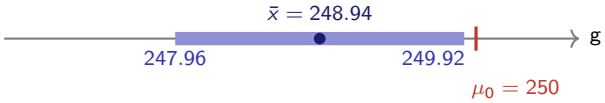
In words: The CI shows *all* μ_0 that you would not reject.

The duality on the number line

Milk: $\mu_0 = 1000$ lies *inside* the CI $\Rightarrow$ **keep** H_0



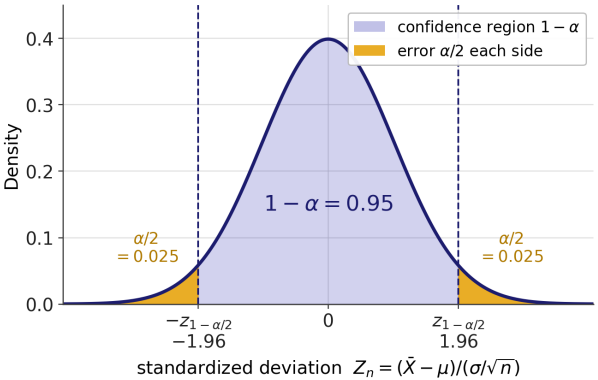
Cream: $\mu_0 = 250$ lies *outside* the CI $\Rightarrow$ **reject** H_0



Key takeaway

μ_0 inside the CI $\Leftrightarrow$ keep H_0; outside $\Leftrightarrow$ reject. (Both at known σ, $\alpha = 0.05$.)

One picture, two readings



- **Distribution reading:** the central blue region holds 95 % of the Z_n distribution (in 95 % of samples Z_n lands here).
- **Test reading:** blue = **acceptance region** (= non-rejection region), orange = **rejection region** (α).
- Test statistic in the orange tail $\Rightarrow$ reject H_0.

"Acceptance region" is conventional for the *non-rejection* region: landing there means we *fail to reject* H_0, not that we accept it as true.

The critical value – the bar you must clear

Plain idea

The **critical value** is a fixed threshold on the test statistic. It splits all possible outcomes into two zones:

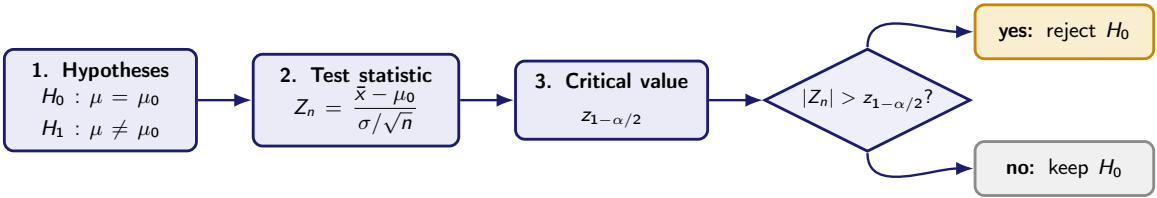
Keep H_0

Statistic *inside* the threshold – the deviation is small enough to pass as ordinary chance.

Reject H_0

Statistic *beyond* the threshold – too extreme to be chance at level α.

The test recipe (known variance)



- **Key move:** plug the hypothesized μ_0 in for the unknown μ – *if* H_0 holds then $Z_n \sim N(0,1)$, so a $|Z_n|$ that would be very unusual for $N(0,1)$ is evidence against H_0.
- $|Z_n| \leq z_{1-\alpha/2}$: acceptance (non-rejection) region; beyond it: rejection region (area α under H_0). Fix α *in advance*.

Milk test with known variance

Data: $\bar{x} = 1001$, $\sigma = 3$, $n = 9$, $\mu_0 = 1000$, $\alpha = 0.05$, $z_{0.975} = 1.96$.

$$Z_n = \frac{1001 - 1000}{3/\sqrt{9}} = \frac{1}{1} = 1.000$$

$$|Z_n| = 1.000 \; < \; 1.96 = z_{0.975}$$

Decision

$H_0 : \mu = 1000$ is **not** rejected – no evidence against a correctly working machine.

In words: Same result as the CI: $\mu_0 = 1000 \in [999.04\,;\,1002.96]$.

Test with unknown variance

Only two changes to the recipe

$$T_n = \frac{\bar{X} - \mu_0}{s/\sqrt{n}} \sim t_{n-1} \text{ under } H_0, \qquad \text{rejection: } |T_n| > t_{1-\alpha/2,\, n-1}.$$

- $\sigma \to s$ (estimated) and $z \to t_{n-1}$.
- **Assumption:** data (approximately) normally distributed *or* $n \geq 30$ (central limit theorem).

Milk test with unknown variance

Data: $\bar{x} = 1001$, $s = 3.7749$, $n = 9$, $\mu_0 = 1000$, $\alpha = 0.05$, $t_{0.975;\,8} = 2.3060$.

$$T_n = \frac{1001 - 1000}{3.7749/\sqrt{9}} = \frac{1}{1.2583} = 0.7947$$

$$|T_n| = 0.7947 \;<\; 2.3060 = t_{0.975;\,8}$$

Decision

$H_0 : \mu = 1000$ is **not** rejected. (More cautious t-quantile 2.31 instead of 1.96 – the result is nevertheless the same here.)

Before you test – the assumptions checklist

1. **Random sample, independent observations** – the X_i are i.i.d. draws from the population. (No clustering, no repeated measurements of the same unit.)
2. **Distribution:** data approximately normal *or* $n \gtrsim 30$ – then the CLT makes $\bar{X}$ approximately normal.
3. **Spread:** σ known $\rightarrow$ z-test; σ estimated by $s \rightarrow$ t-test with $\mathrm{df} = n - 1$.
4. **Question fixed in advance:** μ_0, the direction (one-/two-sided) and α are chosen *before* seeing the results.

Key takeaway

If 1.–2. fail, the stated α is not honest: the 5 % false-alarm guarantee only holds under the model assumptions.

Exercise – cream test

Problem statement

Cream, label 250 g, known variance $\sigma^2 = 4$. Sample $n = 16$, $\bar{x} = 248.9375$. Significance level $\alpha = 0.05$, $\mu_0 = 250$.

Carry out the test – (a) known variance, (b) unknown variance ($s = 2.0156$, $t_{0.975;\,15} = 2.1314$).

Solution (1/2) – cream, known variance

Rule: reject if $|Z_n| > z_{1-\alpha/2}$.

$$Z_n = \frac{248.9375 - 250}{2/\sqrt{16}} = \frac{-1.0625}{0.5} = -2.125$$

$$|Z_n| = 2.125 \; > \; 1.96 = z_{0.975}$$

Decision

$H_0 : \mu = 250$ is **rejected**. There is evidence that the machine does not fill correctly.
(Consistent with the CI: $250 \notin [247.96\,;\,249.92]$.)

Solution (2/2) – cream, unknown variance

$$T_n = \frac{248.9375 - 250}{2.0156/\sqrt{16}} = \frac{-1.0625}{0.5039} = -2.1086$$

$$|T_n| = 2.1086 \; < \; 2.1314 = t_{0.975;\,15}$$

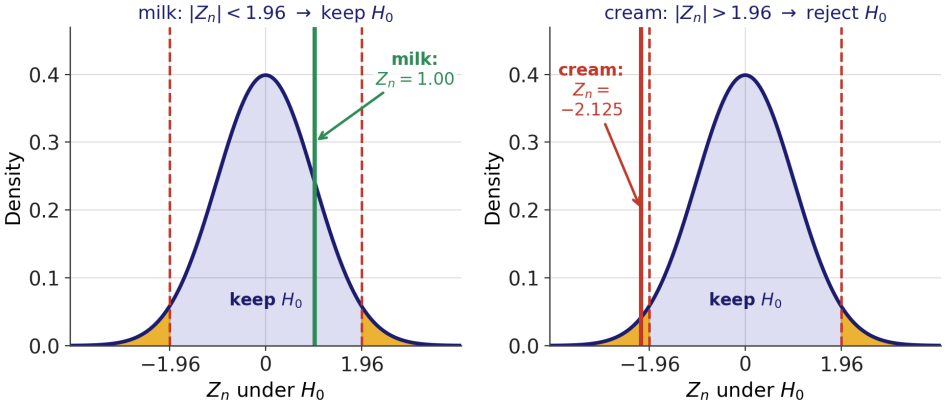
Decision

$H_0 : \mu = 250$ is **not** rejected (just barely!).

Key takeaway

Same data set, a different assumption about σ – a different decision. The t-distribution is more conservative (heavier tails).

Milk vs. cream – the picture behind both decisions



Two-sided test, known σ, $\alpha = 0.05$: the milk statistic ($Z_n = 1.00$) stays *inside* the bars ± 1.96, the cream statistic ($Z_n = -2.125$) lands *beyond* them. The decision is nothing more than: **where does my statistic fall?**

Exercise – reaction time (large sample)

Problem

A training is meant to change the baseline reaction time $\mu_0 = 250\,\text{ms}$. A sample of $n = 50$ gives $\bar{x} = 242$, $s = 30$. Test *two-sided* at $\alpha = 0.05$.

Hint: $t_{0.975;\,49} = 2.0096$ (already close to 1.96).

Solution – reaction time

$$T_n = \frac{242 - 250}{30/\sqrt{50}} = \frac{-8}{4.2426} = -1.8856, \qquad |T_n| = 1.8856 < 2.0096$$

Decision & outlook

Keep H_0 (two-sided): $p = 2\,F_{t,49}(-1.8856) = \mathbf{0.0653} > 0.05$. With $n = 50 \geq 30$ the CLT justifies the t-test. *But* the question may be directional – a one-sided test can decide differently (see the one-sided exercises).

Hypothesis test with R – `t.test`

Command (one-sample t-test)

```
milk ← c(1005, 1001, 997, 1000, 1001, 998, 999, 999, 1009)
t.test(milk, mu = 1000)
```

By default `t.test` performs a **two-sided** t-test with **unknown** variance and directly returns t, df, p-value and CI.

Recap – and one limitation of the recipe

The recipe in one line

Standardize the deviation, compare with the critical value, decide.

Example	Statistic	Bar	Decision	How close?
milk, σ known	$\lvert Z_n \rvert = 1.000$	1.960	keep H_0	clear call
cream, σ known	$\lvert Z_n \rvert = 2.125$	1.960	reject H_0	close call
cream, σ unknown	$\lvert T_n \rvert = 2.109$	2.131	keep H_0	razor-thin

Next section

All three verdicts are a bare *yes/no* – the last column is invisible in them. The p-**value** puts exactly this – the *strength* of the evidence – on a continuous 0–1 scale.

Overview

1 Confidence Intervals

2 Hypothesis Tests

3 The *p*-Value
- Concept
- Calculation
- Exercises

4 One-Sided Tests

5 Types of Error

6 Summary

The *p*-value – intuition

A single question

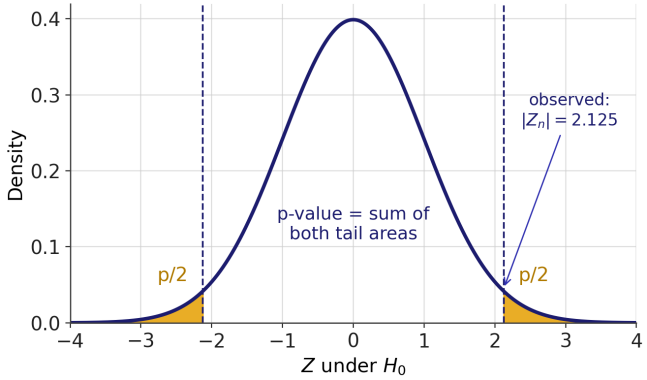
"How **surprising** are my data if H_0 were true?"

- **Small *p*-value** (≈ 0): such an extreme result would be very rare under $H_0 \Rightarrow$ doubt about H_0.
- **Large *p*-value** (≈ 1): the data are unsurprising under $H_0 \Rightarrow$ no reason to doubt it (but *not* positive proof of H_0).

Key takeaway

The *p*-value measures *how surprising* the data are if H_0 holds – the smaller, the stronger the evidence **against** H_0.

The *p*-value – graphically



The (two-sided) *p*-value is the **total area** in both tails that is *at least as extreme* as the observed value.

The *p*-value – definition

Definition

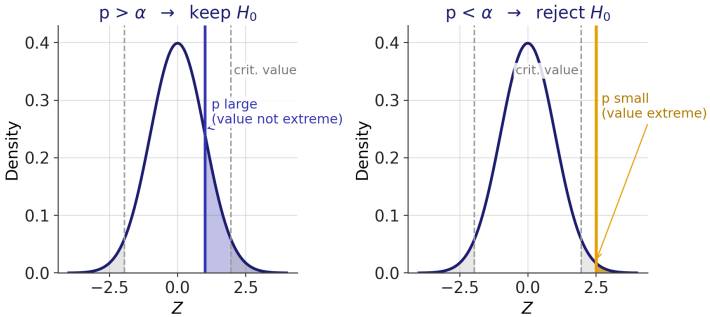
The *p*-value is the probability, under the validity of H_0, of observing a test statistic that is **at least as extreme** as the one actually observed.

- **"At least as extreme" is set by H_1:** two-sided ($\mu \neq \mu_0$) = both tails $|Z| \geq |Z_n|$; one-sided = only the tail in the H_1 direction. Same statistic, different p.
- $0 \leq p \leq 1$; software (e.g. `R`) always reports it.
- **Advantage:** more informative than a bare yes/no – it quantifies the *strength* of the evidence.

Decision rule

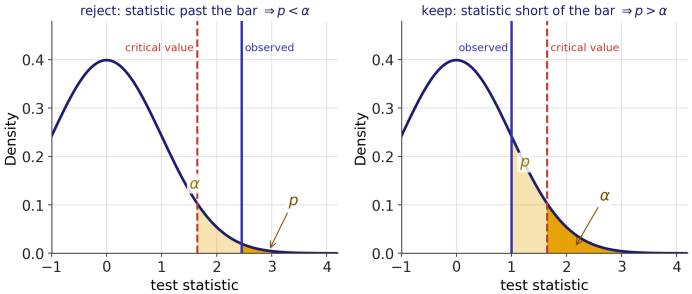
One rule for all tests

$$p < \alpha \implies \text{reject } H_0 \qquad p \geq \alpha \implies \text{keep } H_0$$



- Left: p large, value in the acceptance region.
- Right: p small, value in the rejection region.
- α is the "benchmark".

Two rules, one decision: critical value ↔ *p*-value



- α = tail area beyond the **critical value** (fixed in advance).
- p = tail area beyond your **observed** statistic.
- Statistic past the bar ⇔ its tail is smaller ⇔ $p < \alpha$.

Key takeaway

The two decision rules are **identical**: "test statistic beyond the critical value" and "$p < \alpha$" always agree – the critical value is simply the statistic whose tail area equals α.

Both rules, one verdict – worked example

Cream, two-sided, known σ: $Z_n = -2.125$, $\alpha = 0.05$.

Critical-value rule

$$|Z_n| = 2.125 > 1.96 = z_{0.975}$$

$$\Rightarrow \textbf{reject } H_0.$$

p-value rule

$$p = 2\,\Phi(-2.125) = 0.034$$

$$0.034 < 0.05 \Rightarrow \textbf{reject } H_0.$$

Key takeaway

Same data, same α, **same decision** – because $|Z_n| > z_{0.975}$ and $p < \alpha$ are just two ways of saying the observed value sits past the bar.

Computing the *p*-value (two-sided)

Known variance

$$p = 2 \cdot \Phi\big(-|Z_n|\big) = P\big(|Z| > |Z_n|\big), \quad Z \sim N(0,1)$$

Unknown variance

$$p = 2 \cdot F_{t,n-1}\big(-|T_n|\big)$$

In words: Why double? With $H_1 : \mu \neq \mu_0$ a deviation in *either* direction counts, so "at least as extreme" means *both* tails $|Z| > |Z_n|$; by symmetry each has area $\Phi(-|Z_n|)$ – hence the factor 2. (One-sided: a single tail, no doubling.)

p-value – milk (two variants)

Known variance, $Z_n = 1.000$

$$p = 2 \cdot \Phi(-1.00)$$
$$= 2 \cdot 0.15866 = 0.3173$$

Unknown variance, $T_n = 0.7947$

$$p = 2 \cdot F_{t,8}(-0.7947)$$
$$= 2 \cdot 0.22485 = 0.4497$$

Decision

Both p-values $> 0.05 \Rightarrow H_0 : \mu = 1000$ not rejected. The value 0.4497 matches the `t.test` output.

p-value – cream (two variants)

Known variance, $Z_n = -2.125$

$$\begin{aligned} p &= 2 \cdot \Phi(-2.125) \\ &= 2 \cdot 0.0168 = \mathbf{0.0336} \end{aligned}$$

$0.0336 < 0.05 \Rightarrow$ reject.

Unknown variance, $T_n = -2.1086$

$$\begin{aligned} p &= 2 \cdot F_{t,15}(-2.1086) \\ &= 2 \cdot 0.0261 = \mathbf{0.0522} \end{aligned}$$

$0.0522 > 0.05 \Rightarrow$ keep.

Key takeaway

The same data set can be just *significant* or just *not significant* depending on the assumption about σ.

p-value – typical misconceptions

What the *p*-value is *not*

- **Not** the probability that H_0 is true.
- A large *p*-value does **not prove** H_0 – it only shows *no contradiction*.

Correct – mind the direction

The *p*-value is the probability *of the data (or more extreme)* **assuming** H_0 (chance) holds: $P(\text{data} \mid H_0)$. It *conditions on* chance; it does **not** give the probability that chance was the *cause*, $P(H_0 \mid \text{data})$.

Significant $\neq$ important – the large-n trap

The p-value measures evidence, not size

With enough data, *any* tiny deviation from μ_0 becomes statistically significant – even one that is practically irrelevant.

Milk machine, observed $\bar{x} = 1000.2$ ml ($\mu_0 = 1000$, $\sigma = 3$)

	Z_n	p (two-sided)	verdict at $\alpha = 0.05$
$n = 9$	0.20	0.84	not significant
$n = 10\,000$	6.67	< 0.0001	**highly** significant

Key takeaway

The same irrelevant 0.2 ml deviation – invisible at $n = 9$, "highly significant" at $n = 10\,000$. So report the **effect size and the CI**, not just p.

Exercise – p-value and the choice of α

Problem

A two-sided test with known variance yields $Z_n = 1.85$. Compute the p-value and decide at $\alpha = 0.05$ and at $\alpha = 0.10$.

Hint: $\Phi(1.85) = 0.9678$.

Solution – *p*-value and α

$$p = 2\left(1 - \Phi(1.85)\right) = 2 \cdot 0.0322 = \mathbf{0.0644}$$

- At $\alpha = 0.05$: $0.0644 > 0.05 \Rightarrow$ **keep** H_0.
- At $\alpha = 0.10$: $0.0644 < 0.10 \Rightarrow$ **reject** H_0.

Key takeaway

The same data can be "significant" or not – α must be fixed *in advance*, never chosen to fit the result.

Exercise – interpret the p-value (concept)

Which statements are correct? ($p = 0.03$, $\alpha = 0.05$)

1. "There is a 3 % probability that H_0 is true."
2. "If H_0 were true, data this extreme occur in 3 % of samples."
3. "We reject H_0 at $\alpha = 0.05$."
4. "The effect is large and important."

Solution – interpret the p-value

Answers

1. **Wrong**: p is $P(\text{data} \mid H_0)$, not $P(H_0 \mid \text{data})$.
2. **Correct**: that is exactly the definition.
3. **Correct**: $0.03 < 0.05$.
4. **Wrong**: significance $\neq$ effect size; a tiny effect can be significant with large n.

Key takeaway

A small p-value signals *evidence against* H_0 – not the probability of H_0, and not the size or importance of the effect.

Overview

1 Confidence Intervals

2 Hypothesis Tests

3 The p-Value

4 One-Sided Tests
- Motivation
- Left- and right-sided
- One-sided p-value
- One-sided confidence intervals
- Exercises

5 Types of Error

6 Summary

When only one direction matters

The idea

In a two-sided test *any* departure from μ_0 counts against H_0. Often, though, a deviation is a **problem in only one direction** – the other direction is harmless or even welcome.

Too little in the bottle

For consumer protection only *underfilling* ($\mu < 1000$) is critical – overfilling does not harm the customer.

Unemployment insurance

Only if the rate *exceeds* a threshold do contributions fall short. A lower rate causes no problem.

Key takeaway

The *question* – not the data – decides the direction. The deviation we truly care about defines H_1, and the *whole* α goes on that one side instead of $\alpha/2$ on each.

Business examples – one direction only

You want evidence that ...	H_1	Direction
the new ad campaign *raises* the conversion rate	$\mu > \mu_0$	right-sided
the new process *lowers* the average unit cost	$\mu < \mu_0$	left-sided

In both cases $H_0 : \mu = \mu_0$, as before.

Key takeaway

As in the two-sided case, the claim goes into H_1 – but now its *inequality* also fixes the side on which we reject.

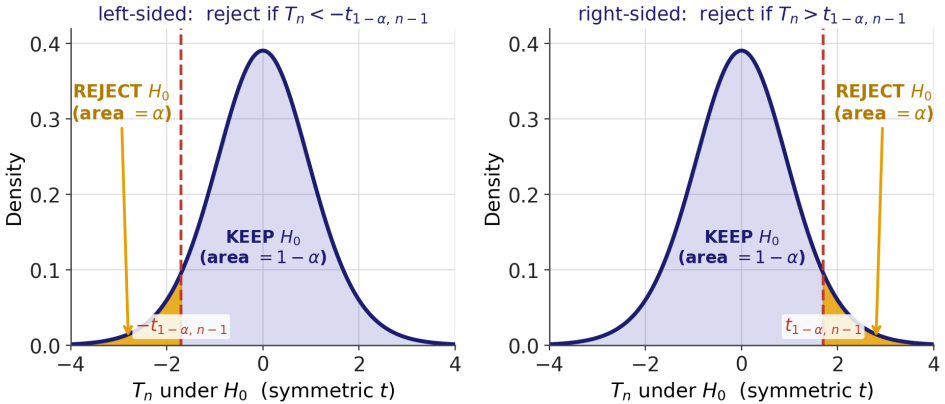
How to find the critical value – 3 steps

1. **Direction** (read from H_1): one-sided or two-sided?
2. **Tail probability:**
   - two-sided $\rightarrow$ *split* α: use the $1 - \alpha/2$ quantile ($\pm$ value),
   - one-sided $\rightarrow$ keep α *whole*: use the $1 - \alpha$ quantile (one value, on the H_1 side).
3. **Which table?** σ known $\rightarrow$ z (normal); σ unknown $\rightarrow$ t with $\mathrm{df} = n - 1$.

$\alpha = 0.05$ – read the numbers off the table

known σ: two-sided $z_{0.975} = 1.96$, one-sided $z_{0.95} = 1.645$.
unknown σ, $n = 12$: two-sided $t_{0.975;\,11} = 2.20$, one-sided $t_{0.95;\,11} = 1.80$.

One-sided critical values – left and right

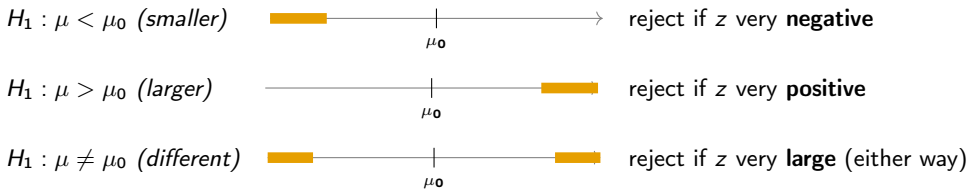


The whole rejection area α sits in *one* tail – left for "too small", right for "too large". The curve is the **symmetric** t-distribution: only the side changes, the shape never does.

The golden rule of one-sided tests

If you remember only one sentence

We reject H_0 when the sample result is **sufficiently extreme in the direction specified by the alternative hypothesis** H_1.



Key takeaway

The inequality in H_1 points straight at the rejection region (orange): "$<$" $\rightarrow$ left tail, "$>$" $\rightarrow$ right tail, "$\neq$" $\rightarrow$ both.

A left-sided test as a lawyer's argument

Your claim ($H_1 : \mu < 10$): "the true mean is *smaller* than 10." The sample mean $\bar{x}$ is your evidence:

$\bar{x} = 8$	$\bar{x} = 9$	$\bar{x} = 10$	$\bar{x} = 11$	$\bar{x} = 12$
strong *for*	some *for*	neutral	against	against

- Below 10 $\Rightarrow$ evidence *for* H_1; if extreme enough, **reject** H_0.
- $\bar{x} = 12$ is far from 10 but in the *wrong* direction – it gives **no** support for "$\mu < 10$".

Key takeaway

The test asks whether the evidence *in the* H_1 *direction* is strong enough to reject H_0 – a big deviation the wrong way counts for nothing.

Critical values of the standard normal $N(0,1)$

Confidence $1-\alpha$	α	**one-sided** $z_{1-\alpha}$	**two-sided** $z_{1-\alpha/2}$
90 %	0.10	1.282	1.645
95 %	0.05	1.645	1.960
98 %	0.02	2.054	2.326
99 %	0.01	2.326	2.576
99.9 %	0.001	3.090	3.291

In words: Pattern: the two-sided value at level α equals the one-sided value at level $\alpha/2$ – e.g. two-sided 95% = 1.96 = one-sided 97.5%.

Critical values of the t-distribution t_{df}

	upper-tail probability (one-sided α | two-sided $\alpha/2$)				
$\mathrm{df} = n-1$	0.10	0.05	0.025	0.01	0.005
1	3.078	6.314	12.706	31.821	63.657
2	1.886	2.920	4.303	6.965	9.925
3	1.638	2.353	3.182	4.541	5.841
4	1.533	2.132	2.776	3.747	4.604
5	1.476	2.015	2.571	3.365	4.032
6	1.440	1.943	2.447	3.143	3.707
7	1.415	1.895	2.365	2.998	3.499
8	1.397	1.860	2.306	2.896	3.355
9	1.383	1.833	2.262	2.821	3.250
10	1.372	1.812	2.228	2.764	3.169
15	1.341	1.753	2.131	2.602	2.947
20	1.325	1.725	2.086	2.528	2.845
30	1.310	1.697	2.042	2.457	2.750
60	1.296	1.671	2.000	2.390	2.660
∞	1.282	1.645	1.960	2.326	2.576

Highlighted: $t_{0.975;\,9} = 2.262$ (two-sided 95%, $n = 10$); the ∞-row is the normal.

How to read the tables – worked lookups

Three questions, every time

(1) one- or two-sided? (2) which α? (3) z or t (df $= n - 1$)?

Normal (σ known):

- two-sided, $\alpha = 0.05 \to$ column $1 - \frac{\alpha}{2} = 0.975$ $\to z = 1.96$.
- one-sided, $\alpha = 0.05 \to 1 - \alpha = 0.95$ $\to z = 1.645$.

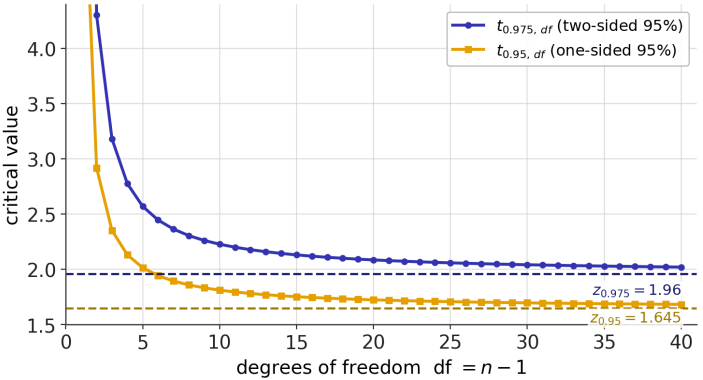
t (σ unknown), $n = 10 \Rightarrow$ df$= 9$:

- two-sided, $\alpha = 0.05 \to$ upper-tail 0.025 $\to t = 2.262$.
- one-sided, $\alpha = 0.05 \to$ upper-tail 0.05 $\to t = 1.833$.

Key takeaway

Two-sided *halves* the tail probability ($\alpha \to \alpha/2$); the t-row is $\mathrm{df} = n - 1$. As df grows, the t-value slides toward z.

Why the t-table ends in the normal



- Small df $\rightarrow$ heavier tails $\rightarrow$ *larger* critical value.
- As df grows, the t-value shrinks toward the z-value.
- By df ≈ 30 the gap is tiny – the ∞ row *is* the normal.

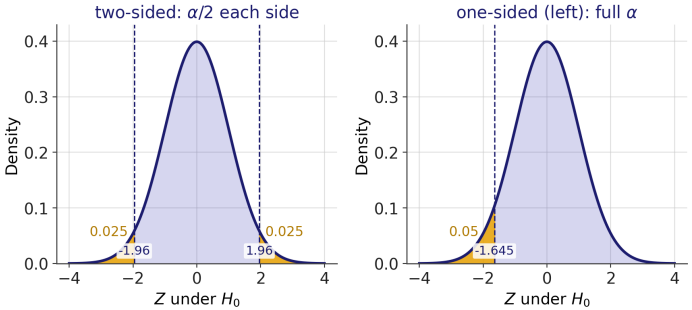
The wording reveals the test direction

Signal words	Hypotheses	Direction
exceed, greater, increased	$H_0 : \mu = \mu_0, \quad H_1 : \mu > \mu_0$	right-sided
fall below, smaller, reduced	$H_0 : \mu = \mu_0, \quad H_1 : \mu < \mu_0$	left-sided

Note

The *question* decides one- or two-sided – not the data. The statement you want to find evidence for goes into H_1.

Why put the whole α on one side?



- Two-sided spends $\alpha/2$ guarding *each* tail – guarding the harmless side is wasted.
- One-sided puts the *whole* α in the relevant tail, so the critical value moves inward ($1.96 \to 1.645$).
- A true deviation in that direction now clears the bar *more easily*.

Key takeaway

Same α, more power – but *only* in the chosen direction. A one-sided test at level α uses the critical value of a two-sided test at level 2α on that side.

The price: you must commit in advance

The direction is part of the hypothesis

Choose one- vs. two-sided **before** looking at the data – based on the research question alone.

- Picking the side *after* seeing which way the data point inflates the true Type I error from α to as much as 2α.
- A one-sided test is **blind** to deviations in the other direction: even a large effect on the "wrong" side never rejects H_0.
- If both directions genuinely matter, use a two-sided test.

In words: One-sided tests buy power with a promise – that the other direction really does not matter.

Left-sided test – consumer protection

Task

A new filling line fills milk (label 1 l). Are the consumer protection requirements met, i.e. is *not too little* being filled? Test with $1 - \alpha = 0.95$.

- $H_0 : \mu = 1000$ vs. $H_1 : \mu < 1000$.
- Only $\mu < 1000$ (too little in the bottle) is critical.

Procedure

Test statistic $T_n = (\bar{X} - \mu_0)/(s/\sqrt{n})$; reject if $T_n < -t_{1-\alpha,\, n-1}$ (known: $Z_n < -z_{1-\alpha}$).
p-value: $p = F_{t,n-1}(T_n)$.

In words: We reject only if $\bar{X}$ falls *clearly below* μ_0 – a value above μ_0 can never be evidence for "too little".

Right-sided test – do not exceed a limit

Task

Advertisement: "Our milk contains **at most** $1\,\text{mg} = 1000\,\mu\text{g}$ of substance K per litre." Is this met? We measure in μg, so $\mu_0 = 1000\,\mu\text{g}$.

- $H_0 : \mu = 1000\,\mu\text{g}$ vs. $H_1 : \mu > 1000\,\mu\text{g}$.
- *Exceeding* the limit is critical.

Procedure

Reject if $T_n > t_{1-\alpha,\,n-1}$ (known: $Z_n > z_{1-\alpha}$).
p-value: $p = 1 - F_{t,n-1}(T_n)$.

In words: Mirror image of the left-sided case: we reject only for values *clearly above* μ_0.

Right-sided test – substance K with numbers

Data (known variance): $\sigma = 90\,\mu$g, $n = 9$, $\bar{x} = 1060\,\mu$g, $\mu_0 = 1000$, $\alpha = 0.05$, $z_{0.95} = 1.645$.

$$Z_n = \frac{1060 - 1000}{90/\sqrt{9}} = \frac{60}{30} = 2.00$$

$$Z_n = 2.00 \;>\; 1.645 = z_{0.95}$$

Decision

$H_0 : \mu = 1000$ is **rejected** ($p = 1 - \Phi(2.00) = 0.0228$). Evidence that the limit is *exceeded* – the advertising claim is not statistically tenable.

Left vs. right-sided – at a glance

	left-sided	**right-sided**
Question	"too *small*?"	"too *large*?"
H_0 / H_1	$\mu = \mu_0$ / $\mu < \mu_0$	$\mu = \mu_0$ / $\mu > \mu_0$
Reject if	$T_n < -t_{1-\alpha,n-1}$	$T_n > t_{1-\alpha,n-1}$
Rejection region	far *left* tail	far *right* tail
p-value	$F_{t,n-1}(T_n)$	$1 - F_{t,n-1}(T_n)$

Some textbooks write the null as $\mu \geq \mu_0$ resp. $\mu \leq \mu_0$ – the test is **identical**, because critical value and p-value are computed at the boundary $\mu = \mu_0$ either way.

Key takeaway

H_1 always points *toward* the rejection region: the inequality in H_1 tells you which tail you reject in.

One-sided p-value – formulas

Left-sided

$$p = F_{t,n-1}(T_n)$$

known: $p = \Phi(Z_n)$

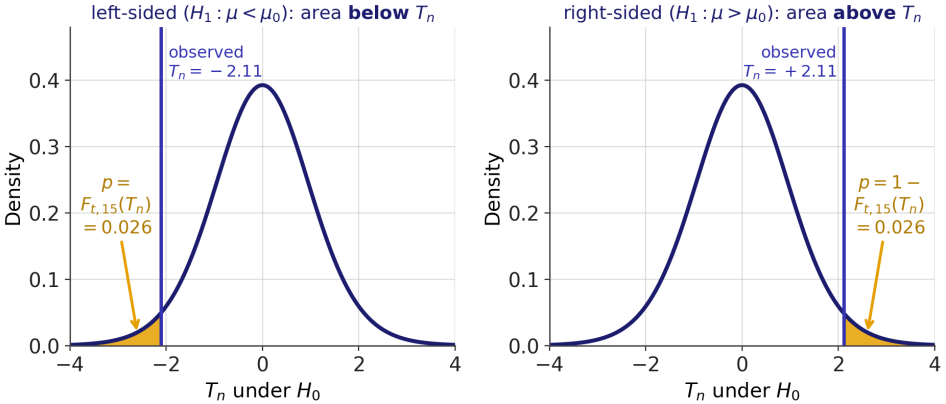
Right-sided

$$p = 1 - F_{t,n-1}(T_n)$$

known: $p = 1 - \Phi(Z_n)$

In words: In a one-sided test we do *not* double – only the one relevant tail area counts.

One-sided p-value – graphically



The p-value is the *single* tail area beyond the observed statistic, **in the direction of** H_1. Left panel: the cream example ($T_n = -2.11$, df $= 15$, $p = 0.026$).

One-sided p-value – mind the sign

Only the H_1-tail counts

The one-sided p-value is the probability, under H_0, of a value *at least as far in the H_1 direction* as the one observed. The opposite tail is irrelevant – a deviation there supports H_0.

Sign check – a big $|z|$ in the wrong direction is *not* significant

Right-sided test with $z = -2.5$: the statistic is on the H_0 side, so $p = 1 - \Phi(-2.5) = 0.994 > 0.5$ – you **never** reject, even though $|z|$ is large. Check the data point in the H_1 direction *before* computing.

In words: Link to two-sided: if the deviation is in the H_1 direction, $p_{\text{one}} = p_{\text{two}}/2$; otherwise $p_{\text{one}} = 1 - p_{\text{two}}/2$.

One-sided p-value – milk ($T_n = 0.7947$, df $= 8$)

Left-sided

$$p = F_{t,8}(0.7947) = 0.7751$$

$> 0.05 \Rightarrow H_0 : \mu = 1000$ kept.

Right-sided

$$p = 1 - 0.7751 = 0.2249$$

$> 0.05 \Rightarrow H_0 : \mu = 1000$ kept.

Interpretation (left)

No evidence that the machine fills *too little* – from a statistical point of view, consumer protection is met.

One-sided p-value – cream (left, unknown var.)

With $T_n = -2.1086$, $\mathrm{df} = 15$:

$$p = F_{t,15}(-2.1086) = 0.0261$$

Decision

$p = 0.0261 < 0.05 \Rightarrow H_0 : \mu = 250$ is **rejected** (in favour of $H_1 : \mu < 250$).

Key takeaway

In contrast to the two-sided test ($p = 0.0522$), the left-sided test is **significant** here – because the whole α lies on the critical side.

One-sided test with R

```r
# left-sided (H1: mu < mu0)
t.test(milk, mu = 1000, alternative = "less")
# right-sided (H1: mu > mu0)
t.test(milk, mu = 1000, alternative = "greater")
```

Cream, left-sided – excerpt

```
t = -2.1086, df = 15, p-value = 0.02606
95 percent confidence interval:  -Inf  249.82
```

The argument `alternative` controls the direction (`"two.sided"`, `"less"`, `"greater"`).

One-sided confidence intervals

- A one-sided test has a one-sided CI – only *one* bound, the other is $\pm\infty$.
- R reports them as $(-\infty;\, x]$ resp. $[x;\, +\infty)$.

Calculation (known variance)

$$\text{left-sided test } (H_1 : \mu < \mu_0): \quad \left(-\infty,\ \bar{X} + z_{1-\alpha}\frac{\sigma}{\sqrt{n}}\right]$$

$$\text{right-sided test } (H_1 : \mu > \mu_0): \quad \left[\bar{X} - z_{1-\alpha}\frac{\sigma}{\sqrt{n}},\ +\infty\right)$$

For unknown variance: replace $z_{1-\alpha}$ by $t_{1-\alpha,\, n-1}$.

Note the one-sided quantile $z_{1-\alpha}$ (not $z_{1-\alpha/2}$) – the whole α sits on one side, exactly as in the test.

One-sided CIs – milk and cream (known)

CI for the *left-sided* test: upper bound ($z_{0.95} = 1.645$):

$$\text{Milk: } 1001 + 1.645 \cdot \frac{3}{\sqrt{9}} = 1002.645$$

$$\text{Cream: } 248.9375 + 1.645 \cdot \frac{2}{\sqrt{16}} = 249.760$$

CI for the *right-sided* test: lower bound:

$$\text{Milk: } 1001 - 1.645 \cdot \frac{3}{\sqrt{9}} = 999.355$$

$$\text{Cream: } 248.9375 - 1.645 \cdot \frac{2}{\sqrt{16}} = 248.115$$

In words: If μ_0 lies beyond the single bound, H_0 is rejected.

Exercise – net weight (left-sided, unknown σ)

Problem

A package must contain *at least* $\mu_0 = 500\,\text{g}$. A sample of $n = 12$ gives $\bar{x} = 496$, $s = 6$. Is there evidence of underfilling? ($\alpha = 0.05$.)

Hint: $t_{0.95;\,11} = 1.7959$.

Solution – net weight

Underfilling is critical $\Rightarrow H_0 : \mu = 500$ vs. $H_1 : \mu < 500$ (left-sided). Reject if $T_n < -t_{0.95;\,11} = -1.7959$.

$$T_n = \frac{496 - 500}{6/\sqrt{12}} = \frac{-4}{1.7321} = -2.3094 < -1.7959$$

Decision

Reject H_0: evidence of underfilling. $p = F_{t,11}(-2.3094) = \mathbf{0.0207} < 0.05$.

Exercise – one- vs. two-sided on the same data

Problem

A coating must *exceed* a target thickness. A test gives $T_n = 1.80$ with df $= 15$. Compare the right-sided and the two-sided decision at $\alpha = 0.05$.

Hint: $1 - F_{t,15}(1.80) = 0.0460$.

Solution – one- vs. two-sided

$$\text{right-sided: } p = 1 - F_{t,15}(1.80) = \mathbf{0.0460} < 0.05 \;\Rightarrow\; \text{reject}$$
$$\text{two-sided: } p = 2 \cdot 0.0460 = \mathbf{0.0920} > 0.05 \;\Rightarrow\; \text{keep}$$

Key takeaway

Identical data, opposite decisions. The one-sided test is more powerful *in the chosen direction* – exactly why the direction must be fixed before seeing the data.

Overview

1 Confidence Intervals

2 Hypothesis Tests

3 The p-Value

4 One-Sided Tests

5 Types of Error
- The two types of error
- α, β and power
- Exercises

6 Summary

Two ways to be wrong

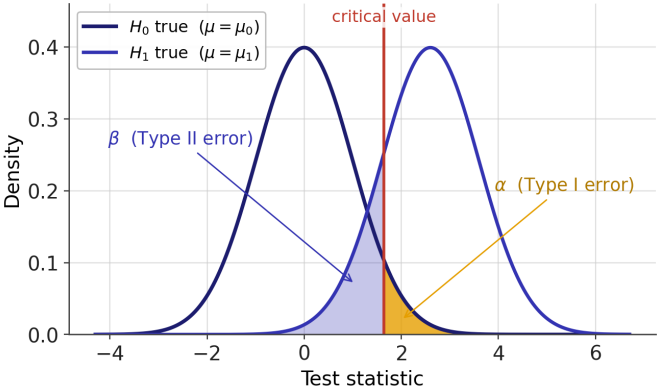
Intuition

Like a smoke detector, we can be wrong in two ways: **false alarm** or **missed fire**.

Decision	H_0 **true**	H_0 **false**
reject H_0	Type I error (α)	correct (power $1-\beta$)
keep H_0	correct ($1-\alpha$)	Type II error (β)

Type I error = false alarm (reject although H_0 is true). **Type II error** = miss (keep although H_1 is true).

α and β in a picture



- The two curves are $\bar{X}$ in each "world": left centred at μ_0 (if H_0 holds), right at the true μ_1 (if H_1); the red line is the critical value.
- α: H_0 true, but beyond the critical value.
- β: H_1 true, but on this side.
- The distributions overlap – both errors cannot be 0 at once.

Type I error (α) – controllable

Definition

$$P(\text{reject } H_0 \mid H_0 \text{ true}) = \alpha$$

- **Directly controllable** through the choice of α (typically 0.05).
- In 5 % of cases a *true* H_0 is wrongly rejected.

Milk

$H_0 : \mu = 1000$ is true, yet the test rejects (false alarm) – probability $= \alpha$.

Type II error (β) – only indirectly influenced

Definition (at a specific alternative μ_1)

$$P(\text{keep } H_0 \mid \mu = \mu_1) = \beta$$

- No *direct* lever; β depends on:
  - α, sample size n, spread σ^2,
  - true deviation $|\mu - \mu_0|$ (effect size).
- At a fixed μ_1: smaller $\alpha \Rightarrow$ higher bar $\Rightarrow$ more misses (β *rises*); larger $n \Rightarrow$ smaller SE $\Rightarrow$ β *falls*.

Key takeaway

Power $= 1 - \beta$: the ability to actually reject a *false* H_0.

Only one error is controllable

Question: Which error can we only make when we *reject* H_0?

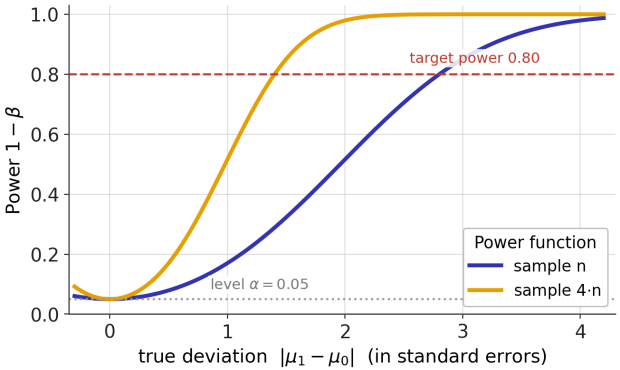
Answer

The Type I error – and that is exactly the one we control via α.

Consequence for the choice of hypotheses

This is why the "worse" alternative is formulated as H_1: the feared error becomes the controllable Type I error.

Trade-off and power



- $\alpha\downarrow \Rightarrow \beta\uparrow$ (trade-off).
- Larger n raises power at every $\mu \neq \mu_0$ (at μ_0 it stays at α) – more power at the same level.
- Power rises with the true deviation.

Practice

Typically $\alpha = 0.05$ and desired power $1 - \beta \geq 0.80$. A **power analysis** provides the required n.

Computing β concretely – milk

Setup: $H_0 : \mu = 1000$ vs. $H_1 : \mu \neq 1000$, $\sigma = 3$, $n = 9$ $(\mathrm{SE} = 1)$, $\alpha = 0.05$.

Step 1 – turn "$|Z_n| \leq 1.96$" into bounds on $\bar{X}$:
$\mu_0 \pm 1.96 \cdot \mathrm{SE} = 1000 \pm 1.96 = [998.04\,;\,1001.96]$.

Step 2 – the truth is $\mu_1 = 1002$; find how often $\bar{X}$ still lands in that region, *standardizing with* μ_1:

$$\begin{aligned}\beta &= P\big(998.04 \leq \bar{X} \leq 1001.96 \;\big|\; \mu = 1002\big) \\ &= \Phi\Big(\tfrac{1001.96-1002}{1}\Big) - \Phi\Big(\tfrac{998.04-1002}{1}\Big) = 0.484\end{aligned}$$

Result

Power $= 1 - \beta = 0.516$: a true deviation of 2 ml is detected with $n = 9$ in only just over half the cases.

Four levers for more power

More sample size helps decisively. Same situation with $n = 36$ instead of 9:

$$\text{SE} = \frac{3}{\sqrt{36}} = 0.5 \;\Rightarrow\; \beta = 0.021, \quad \text{power} = 0.979.$$

Levers of power $1 - \beta$

1. **Sample size** $n \uparrow$ (strongest lever).
2. **Effect size** $|\mu_1 - \mu_0|$ (given).
3. **Spread** $\sigma \downarrow$ (e.g. measure more precisely).
4. **Level** $\alpha \uparrow$ (buys more Type I error).

Key takeaway

Quadrupling n from 9 to 36 raises the power from 0.52 to 0.98.

Exercise – compute β and power (two-sided)

Problem

Test $H_0 : \mu = 100$ vs. $H_1 : \mu \neq 100$, known $\sigma = 10$, $n = 25$, $\alpha = 0.05$. The true mean is actually $\mu_1 = 105$. Find the Type II error β and the power.

Step 1: the acceptance region for $\bar{X}$ under H_0.

Solution – β and power (two-sided)

SE $= 10/\sqrt{25} = 2$. Acceptance region: $100 \pm 1.96 \cdot 2 = [96.08\,;\,103.92]$.

$$\begin{aligned}\beta &= P(96.08 \le \bar{X} \le 103.92 \mid \mu = 105) \\ &= \Phi\Big(\tfrac{103.92-105}{2}\Big) - \Phi\Big(\tfrac{96.08-105}{2}\Big) = \Phi(-0.54) - \Phi(-4.46) \\ &= 0.2946 - 0.0000 = \mathbf{0.2946}\end{aligned}$$

Power

$1 - \beta = \mathbf{0.7054}$: a true shift of 5 units is detected in about 71 % of samples.

Exercise – power of a one-sided test

Problem

Now use a *right-sided* test, $H_1 : \mu > 100$, with $\sigma = 10$, $n = 25$, $\alpha = 0.05$, true $\mu_1 = 105$. Find the power and compare with the two-sided value 0.7054.

Solution – one-sided power

SE $= 2$. Critical mean: $\bar{x}_{\text{crit}} = 100 + 1.645 \cdot 2 = 103.29$.

$$1 - \beta = P(\bar{X} > 103.29 \mid \mu = 105) = 1 - \Phi\left(\tfrac{103.29 - 105}{2}\right) = 1 - \Phi(-0.855) = \mathbf{0.8037}$$

Comparison

Power 0.804 vs. 0.705 two-sided – the one-sided test is *more powerful* for the same n and α, because the whole α sits on the relevant side.

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Summary 1 – confidence intervals

Known variance σ $(Z_n \sim N(0,1))$

$$\mathrm{CI}_{1-\alpha} = \left[\bar{X} - z_{1-\alpha/2}\frac{\sigma}{\sqrt{n}},\ \bar{X} + z_{1-\alpha/2}\frac{\sigma}{\sqrt{n}}\right]$$

Unknown variance $(T_n \sim t_{n-1})$

$$\mathrm{CI}_{1-\alpha} = \left[\bar{X} - t_{1-\alpha/2,\,n-1}\frac{s}{\sqrt{n}},\ \bar{X} + t_{1-\alpha/2,\,n-1}\frac{s}{\sqrt{n}}\right]$$

Summary 2 – hypothesis test (two-sided)

$H_0 : \mu = \mu_0$ vs. $H_1 : \mu \neq \mu_0$

Test statistics

$$Z_n = \frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}} \sim N(0,1), \qquad T_n = \frac{\bar{X} - \mu_0}{s/\sqrt{n}} \sim t_{n-1}$$

Decision

Reject if $|Z_n| > z_{1-\alpha/2}$ resp. $|T_n| > t_{1-\alpha/2,\, n-1}$.

Summary 3 – the p-value

One idea behind every formula

p = probability, *if* H_0 *were true*, of a value **at least as extreme as observed** – in the direction(s) that H_1 allows.

If H_1 is...	**add up the...**	p **(unknown σ)**
$\mu \neq \mu_0$	*both* tails beyond $\lvert T_n \rvert$	$2\,F_{t,n-1}(-\lvert T_n \rvert)$
$\mu < \mu_0$ (left)	*lower* tail below T_n	$F_{t,n-1}(T_n)$
$\mu > \mu_0$ (right)	*upper* tail above T_n	$1 - F_{t,n-1}(T_n)$

Known σ? Just replace $T_n \to Z_n$ and $F_{t,n-1} \to \Phi$. **Decide:** $p < \alpha \Rightarrow$ reject H_0.

Summary 4 – one-sided tests & types of error

One-sided tests (unknown variance)

left: $H_1 : \mu < \mu_0$ reject if $T_n < -t_{1-\alpha,\, n-1}$

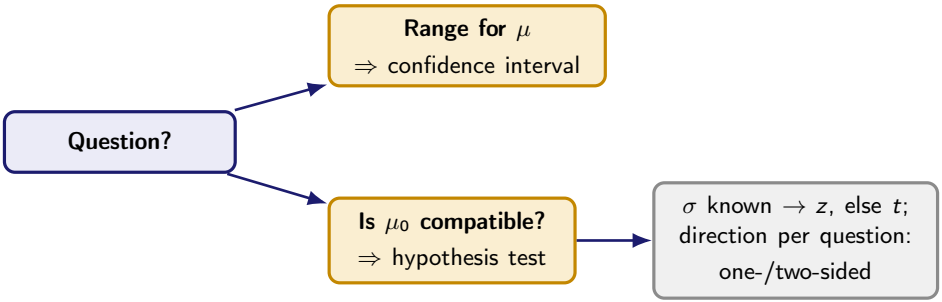
right: $H_1 : \mu > \mu_0$ reject if $T_n > t_{1-\alpha,\, n-1}$

Types of error

$P(\text{reject } H_0 \mid \mu = \mu_0) = \alpha$ $P(\text{keep } H_0 \mid \mu = \mu_1) = \beta$

α is *controllable*; β falls with $n \uparrow$ and $\alpha \uparrow$; power $= 1 - \beta$.

Which procedure? – decision tree



The same logic – *standardize*, then ask whether the deviation is extreme *in the H_1 direction* – underlies essentially every classical test: z, t, proportion, regression coefficient, ANOVA F.

Key takeaway

First clarify the *question*, then choose the tool.

Five classic pitfalls – final checklist

Forbidden sentence ×	**Correct version** ✓
"p is the probability that H_0 is true."	$p = P(\text{data at least this extreme} \mid H_0)$.
"μ lies in *this* CI with 95 % probability."	The *procedure* covers μ in 95 % of samples.
"$p > \alpha$, so H_0 is proven/accepted."	No contradiction found – H_0 is *kept*, not proven.
"The data point left, so I test left-sided."	Direction and α are fixed *before* the data.
"Significant, hence large and important."	Significance $\neq$ size: judge magnitude by the CI.

Outlook: Part 4 – Regression

What we have so far

Confidence intervals and tests for **one** population mean μ: estimate, standard error, critical value – and the p-value.

Next part

We study the **relationship between variables**. The regression slope $\hat{\beta}_1$ is again an estimator with a standard error, so the *same* recipe gives its t-test, its confidence interval and its p-value:

$$T = \frac{\text{estimate} - \text{hypothesized value}}{\text{standard error}}.$$

Key takeaway

Everything in this part carries over one-to-one to the coefficients of a regression model.